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Employee Retention Prediction

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- Course: Data Science & AI
- Project: Employee Job change Prediction using Machine Learning

Agenda:

- Introduction
- Problem Statement
- Dataset Overview
- Methodology
- Model Building & Results
- Model Comparison
- Real-World Application
- Conclusion

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Introduction:

- Employee retention is a major challenge for organizations.
- High attrition increases hiring and training costs.
- Machine learning enables prediction of employee job change behavior.
- This project applies data science techniques to HR analytics.

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Problem Statement:

- Organizations invest heavily in employee training
- Some employees leave after training
- This leads to financial and productivity loss

Objective:

Predict whether an employee is likely to look for a job change

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Dataset Overview:

HR dataset containing employee demographic and professional details.
It includes education, experience, company size, city development index and training hours.

Target Variable:

0- Not looking for job change
1- Looking for job change

Exploratory Data Analysis (EDA):

EDA was performed to understand employee behavior and key factors affecting job change.
Experience, training hours, and company environment were found to strongly influence attrition and guided feature selection.

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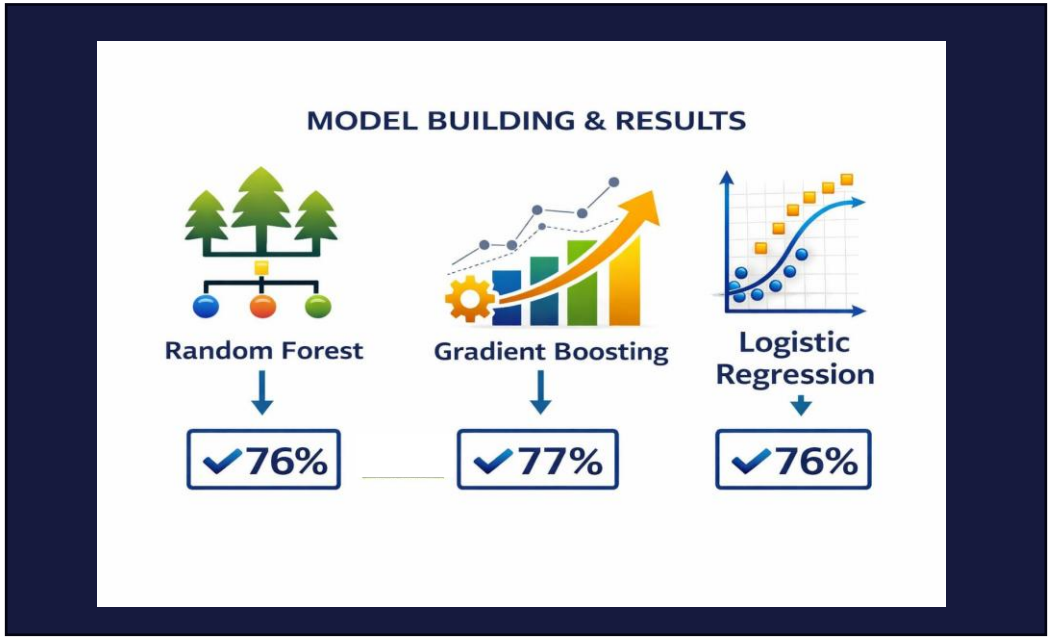
Data Preprocessing:

- Handled missing values using median and mode
- Encoded categorical variables into numeric values
- Prepared clean data for machine learning models

Model Building & Result:

❑Algorithms Used:

- Random Forest – Accuracy ~76.67%
- Gradient Boosting – Accuracy ~77.71%
- Logistic Regression – Accuracy ~76.56%



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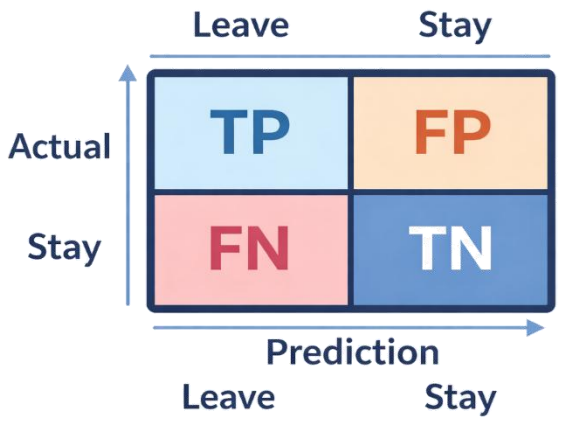
Performance Evaluation Metrics

Confusion Matrix:

- It shows how many predictions are correct and incorrect by comparing actual vs predicted classes in a table format.
- It represents True Positive, True Negative, False Positive, and False Negative.
- It helps understand where the model is making mistakes.



Accuracy:

- It measures how many total predictions made by the model are correct.
- Out of all employees, how many were correctly predicted (leave or stay).
- Gives overall performance of the model.
- Best metric when dataset is balanced.

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Precision:

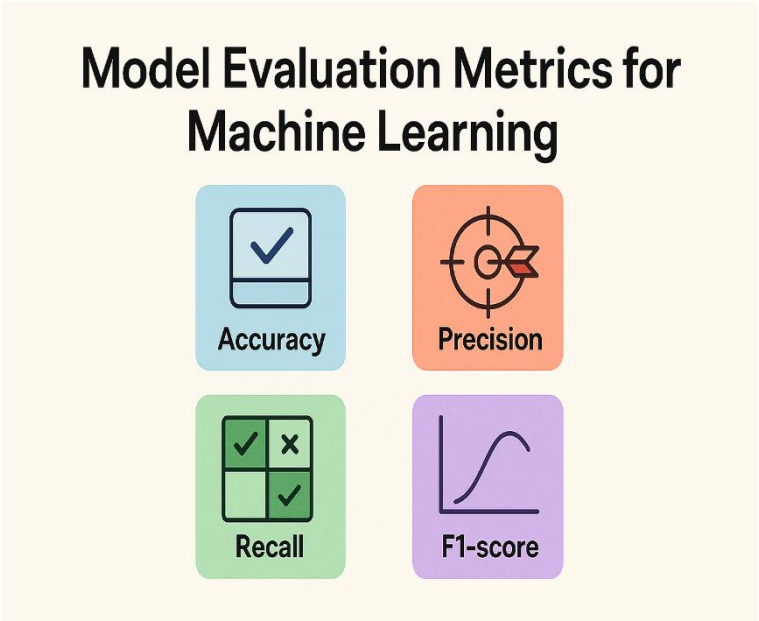
- Out of all employees predicted to change jobs, how many actually changed.
- Useful to avoid wrong warnings to HR.

Recall:

- Out of all employees who actually changed jobs, how many were detected.
- Important to reduce missing high-risk employees.

F1 Score:

- It Balances Precision and Recall into a single metric.
- Best metric when dataset is imbalanced.



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Algorithms:

Random Forest:

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their results to make accurate predictions. It reduces overfitting and works well with large and complex datasets.

Gradient Boosting:

Gradient Boosting is an advanced ensemble technique that builds models sequentially, where each new model corrects the errors of the previous one. This results in higher accuracy and better performance on structured data.

Logistic Regression:

Logistic Regression is a simple algorithm used for binary classification problems. It estimates the probability of an outcome using a logistic function. The model is easy to interpret and works well with linearly separable data.

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Result Interpretation:

Gradient Boosting achieved slightly higher accuracy (~77.71%) compared to Random Forest (~76.67%) and Logistic Regression (~76.56%) making it the preferred model for predicting employee job change.

HR Insights:

- Identifies employees at risk of leaving
- Enables proactive retention strategies
- Reduces hiring and training costs
- Supports data-driven HR decisions

Gradient Boosting

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Real-World Business Impact:

- Identifies employees at risk of leaving
- Enables proactive retention strategies
- Reduces hiring and training costs
- Supports data-driven HR decisions

Conclusion:

The project successfully predicts employee job change behavior.
Machine learning improves workforce planning.
The solution is scalable for real-world HR analytics.

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Questions ?

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Thank You!